

# Configuration Manual

Physics-Informed HVAC Energy Prediction under Incomplete Sensor Data

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September 2026

## 1 Purpose

This manual explains how to set up the environment, obtain the data and reproduce every table and figure of the report. All commands are run from the project folder `satish-thesis/`.

## 2 Hardware

| Item                      | Used for the report                 | Minimum                                     |
|---------------------------|-------------------------------------|---------------------------------------------|
| Processor                 | Apple silicon, 8+ cores             | any 64-bit x86 or ARM CPU                   |
| Accelerator               | Apple MPS (automatic)               | none (CPU works); CUDA is used when present |
| Memory                    | 16 GB                               | 8 GB                                        |
| Disk                      | 1 GB (400 MB for optional raw data) | 200 MB                                      |
| Run time, full experiment | about 45 minutes                    | about 90 minutes on 4 CPU cores             |

## 3 Software

| Package                          | Version                                       |
|----------------------------------|-----------------------------------------------|
| Python                           | 3.13.9 (3.10 or newer works)                  |
| numpy / pandas / scipy           | 2.4.6 / 2.3.3 / 1.18.0                        |
| scikit-learn                     | 1.5.1                                         |
| xgboost                          | 3.2.0                                         |
| torch                            | 2.11.0                                        |
| matplotlib                       | 3.10.6                                        |
| pytest, jupyter / nbconvert      | any recent version                            |
| LaTeX (optional, for the report) | TeX Live or TinyTeX with <code>latexmk</code> |

## 4 Set-up

### 4.1 macOS or Linux

```
git clone https://github.com/Tarunchintakunta/Thesis-2.0.git
cd Thesis-2.0/satish-thesis
```

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

## 4.2 Windows (PowerShell)

```
git clone https://github.com/Tarunchintakunta/Thesis-2.0.git
cd Thesis-2.0\satish-thesis
py -3 -m venv .venv
.venv\Scripts\Activate.ps1
pip install -r requirements.txt
```

For an NVIDIA GPU on Windows or Linux, install the CUDA build of PyTorch from <https://pytorch.org> before the other requirements.

## 4.3 Google Colab

Open `notebooks/physics_informed_hvac.ipynb` in Colab (File, Open notebook, GitHub) and choose a GPU runtime if available. The first cell clones only this project with a sparse checkout and installs missing packages. No other step is needed.

# 5 Data

The hourly table used by all experiments is bundled as `data/bldg59_hourly.csv.gz`, so the steps below are optional. To rebuild it from the raw LBNL Building 59 files (Dryad, doi:10.7941/D1N33Q):

1. Create a Kaggle API token (Kaggle, Settings, Create New Token) and save it as `~/.kaggle/kaggle.json`. Dryad itself blocks scripted downloads, so a Kaggle mirror of the same files is used.
2. Run the download script and the preparation step:

```
bash scripts/download_data.sh
python src/prepare_data.py
```

This writes `data/processed/bldg59_hourly.csv`, the compressed copy, `results/data_dictionary.csv` and `results/data_profile.json`. Raw files stay in `data/raw/`, which git ignores.

# 6 Running the project

## 6.1 Tests

```
python -m pytest tests -q
```

Expected output: 19 passed.

## 6.2 Smoke run (2–3 minutes)

```
python src/experiment.py --quick
```

It trains every model for two epochs with one seed and writes files ending in `_quick`. Use it to check the installation.

## 6.3 Full experiment

```
python src/experiment.py
python src/analysis.py
```

experiment.py runs, for each task (estimate, then forecast), the tree family and then the neural family in separate processes. Options:

- -task estimate|forecast runs one task only;
- -family tree|nn runs one model family only;
- -stage main|imputation runs only the main grid or only the imputation sensitivity study.

Selected hyperparameters are cached in results/chosen\_<task>\_<family>.json; delete these files to re-tune. analysis.py writes tables to results/tables/ (CSV and LaTeX) and figures to figures/.

## 6.4 Notebook

```
jupyter notebook notebooks/physics_informed_hvac.ipynb
```

With RUN\_FULL=False (default) it runs the tests, the smoke run and the analysis on the stored results (about 5 minutes). With RUN\_FULL=True it re-runs the full experiment first. To execute it without a browser:

```
jupyter nbconvert --to notebook --execute --inplace notebooks/physics_informed_hvac.
    ipynb
```

## 6.5 Report

```
cd report
latexmk -pdf thesis.tex
```

# 7 Outputs

| File                                    | Content                                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| results/metrics_<task>_<family>.csv     | One row per model, seed, pattern and rate: MAE, RMSE, MAPE, $R^2$ , NRMSE, negative-prediction share, temperature RMSE and RC residual. |
| results/predictions_<task>_<family>.npz | Test predictions for every model, seed and condition, targets and timestamps.                                                           |
| results/tuning_<task>_<family>.csv      | Every tuning trial and its validation RMSE.                                                                                             |
| results/physics_coefs_<task>_nn.csv     | Learned RC coefficients per seed.                                                                                                       |
| results/plausibility_*.csv, cost_*.csv  | Outdoor-temperature response test; training and inference time.                                                                         |
| results/imputation_estimate_*.csv       | Mean and CART imputation runs.                                                                                                          |
| results/tables/, figures/               | Everything shown in the report.                                                                                                         |

# 8 Troubleshooting

- Segmentation fault or a hang on macOS when XGBoost and PyTorch are used in one Python process. Both ship an OpenMP runtime. Always run src/experiment

.py, which keeps them in separate processes; do not import `xgboost` in the same session as a PyTorch training loop.

- **MPS not used.** Check `python -c "import torch; print(torch.backends.mps.is_available())"`; it needs macOS 12.3+ and an arm64 Python.
- **Small numeric differences between runs on GPU.** MPS and CUDA kernels are not bit-deterministic; differences are well inside the seed-to-seed spread reported in the tables.
- **kaggle: command not found.** Install it with `pip install kaggle`, or skip the download and use the bundled table.